

Name: _____
 Class: _____
 Date: _____

Mr. Rawson • IB DP Physics

Simple Harmonic Motion 7 – Practice IB Questions

Paper 1A and Paper 2 style

Before you start

Have your **calculator in radians** and your **IB data booklet** open. In every calculation, write the equation first, then substitute, then give the answer with a unit. **Check:** $\sin(\pi/2) = 1$ on your calculator.

The command terms on this sheet, as the IB defines them

State	Give a specific name, value or other brief answer without explanation or calculation.
Calculate	Obtain a numerical answer showing the relevant stages in the working.
Show	Give the steps in a calculation or derivation. In <i>show that</i> , the answer is given: the marks are for the working, so give one more significant figure than the value shown.
Determine	Obtain the only possible answer.
Explain	Give a detailed account including reasons or causes.

SL

Everyone answers this section.

Multiple choice (Paper 1A style: circle one letter)

Question 1

1 mark

Relevant equations: $\omega = 2\pi f$

An object oscillates with simple harmonic motion at a frequency of 2.5 Hz. What is the angular frequency of the oscillation?

- A. 0.40 rad/s B. 2.5 rad/s C. 7.9 rad/s D. 16 rad/s

Question 2

1 mark

Relevant equations: $T = 2\pi\sqrt{\frac{m}{k}}$ $v_{\max} = \omega x_0$ $a_{\max} = \omega^2 x_0$

A mass on a spring oscillates with simple harmonic motion. The amplitude is doubled. What happens to the period T , the maximum speed $v_{\max}$ and the maximum acceleration $a_{\max}$?

	T	$v_{\max}$	$a_{\max}$
A.	unchanged	unchanged	unchanged
B.	unchanged	doubled	doubled
C.	doubled	unchanged	unchanged
D.	doubled	doubled	quadrupled

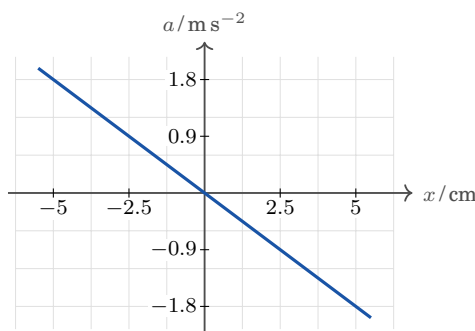
Question 3**1 mark****Relevant equations:** $v_{\max} = \omega x_0$ $a_{\max} = \omega^2 x_0$ $T = \frac{2\pi}{\omega}$

An oscillator in simple harmonic motion has a maximum speed of 0.60 m/s and a maximum acceleration of 3.0 m/s². What is its period?

- A.** 0.20 s **B.** 1.3 s **C.** 5.0 s **D.** 7.9 s

Question 4**1 mark****Relevant equations:** $a = -\omega^2 x$ $\omega = 2\pi f$

The graph shows how the acceleration a of an oscillating object varies with its displacement x .



What is the frequency of the oscillation?

- A.** 0.95 Hz **B.** 5.7 Hz **C.** 6.0 Hz **D.** 36 Hz

Structured questions (Paper 2 style)**Question 5****6 marks****Relevant equations:** $\omega = \frac{2\pi}{T}$ $v_{\max} = \omega x_0$ $a_{\max} = \omega^2 x_0$ $a = -\omega^2 x$

A car drives over a bump. Afterwards, the car body moves up and down on its springs with simple harmonic motion of period 0.80 s and amplitude 4.0 cm.

- (a)** Calculate the angular frequency of the oscillation.

[1]

- (b)** Calculate the maximum speed of the car body.

[1]

- (c) Calculate the maximum acceleration of the car body, and state where in the oscillation it occurs.

[2]

- (d) The motion of the car body is described by $a = -\omega^2 x$. State **two** features of the acceleration that this equation describes.

[2]

Question 6

6 marks

Relevant equations: $T = 2\pi\sqrt{\frac{l}{g}}$ $\omega = \frac{2\pi}{T}$ $v_{\max} = \omega x_0$

A child sits on a playground swing. The swing behaves as a simple pendulum of length 2.5 m.

- (a) Show that the period of the swing is about 3.2 s.

[2]

- (b) The seat moves 0.60 m either side of its equilibrium position. Calculate the maximum speed of the seat.

[2]

- (c) A heavier child sits on the same swing. State and explain the effect on the period.

[2]

Relevant equations: $\omega = \sqrt{\frac{k}{m}}$ $T = \frac{2\pi}{\omega}$ $a = -\omega^2 x$

A trolley of mass 0.40 kg sits between two springs on a level track. When it is displaced by x from its equilibrium position, the resultant force on it is $F = -36x$, with F in newtons and x in metres.

- (a) Explain why the trolley performs simple harmonic motion when it is released.

[2]

- (b) Calculate the period of the oscillation.

[2]

HL students: answer this section as well. It uses the HL equations from your data booklet.

Multiple choice (Paper 1A style: circle one letter)

Question 8

1 mark

Relevant equations: $x = x_0 \sin \omega t$ $\omega = \frac{2\pi}{T}$

An object moves in simple harmonic motion with period T and amplitude x_0 . It passes through the equilibrium position at $t = 0$. What is the shortest time for it to reach a displacement of $\frac{1}{2}x_0$?

- A. $\frac{T}{12}$ B. $\frac{T}{8}$ C. $\frac{T}{6}$ D. $\frac{T}{4}$

Question 9

1 mark

Relevant equations: $x = x_0 \sin(\omega t + \phi)$ $v = \omega x_0 \cos(\omega t + \phi)$

The displacement of an oscillator is given by $x = x_0 \sin(\omega t + \phi)$. At $t = 0$ the oscillator passes through the equilibrium position moving in the negative direction. What is ϕ ?

- A. 0 B. $\frac{\pi}{2}$ C. π D. $\frac{3\pi}{2}$

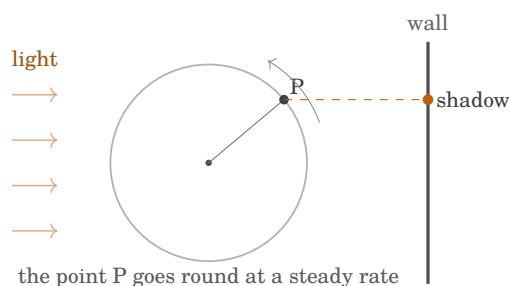
Structured questions (Paper 2 style)

Question 10

7 marks

Relevant equations: $\omega = 2\pi f$ $x = x_0 \sin \omega t$ $v = \omega x_0 \cos \omega t$ $v = \pm \omega \sqrt{x_0^2 - x^2}$

A small peg P is fixed to the edge of a turntable, 0.12 m from its centre. The turntable rotates at a steady 45 revolutions per minute. Light shines from the side, so the shadow of P on a wall moves up and down with simple harmonic motion.



- (a) Show that the angular frequency of the shadow's motion is about 4.7 rad/s.

[1]

- (b) At $t = 0$ the shadow passes through its equilibrium position, moving upwards. Calculate the displacement of the shadow at $t = 0.20$ s.

[2]

- (c) Calculate the velocity of the shadow at $t = 0.20$ s.

[2]

- (d) Use your answer to (b) to confirm your answer to (c).

[1]

- (e) A second peg is fixed to the edge of the same turntable, a quarter of a turn ahead of P. State the phase difference between the motions of the two shadows.

[1]
